Scoping the Next Generation of Tuna Stock Assessment Software (P123)

External Tagging Analysis and Assessment Model Design

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SPC PAW Meeting
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Overview

External Tagging Analysis *method being developed by DTU scientists, related but different from SEAPODYM*

Assessment Model Design *SPC – IATTC – NOAA – Quantifish collaboration, RTMB estimation platform, time-varying selectivity*

SC22 Project Proposal *budget, work plan, timeline*

External tagging analysis

Method being developed by DTU scientists, in collaboration with SPC and IATTC

The current analysis focuses on WCPO skipjack tags, with the intention to analyze yellowfin and bigeye tags later

Initial workshop in 2025

- Tobias Mildenberger, Anders Nielsen (DTU)
- Inna Senina, Joe Scutt Phillips, John Hampton, Arni Magnusson (SPC)
- Mark Maunder, Rujia Bi (IATTC)

Currently securing funding for this work to proceed

- small amount for 2026 (from P123 budget)
- larger amount for 2027 (from upcoming Software Development Project budget)
- method development could be completed in 2028 *for that year's SKJ assessment*

External tagging analysis

Related but different from SEAPODYM

DTU Movement Model

Uses all tags, also those that are not recaptured

Designed to produce abundance indices

Software maintained by Tobias Mildenberger (DTU) using RTMB

SEAPODYM

Only uses recaptured tags

Designed to estimate fish movement

Software maintained by Inna Senina (SPC) using AUTODIF C++ libraries

Assessment model design

CAPAM 2025

Mini-conference showcasing cutting-edge research across many research domains in stock assessment modelling

https://www.populationassessment.com/tuna/next-gen-model/workshop

Priority features

- ▶ Features that are currently used in WCPFC tuna stock assessments
- \+ Time-varying selectivity, using random effects
- – No tagging analysis inside the model

Development paradigm and first steps

Continuous and dynamic

- Design decisions will be made based on findings during development
- Mark Maunder contributes to model design coordination

OPAL project

- Nicholas Ducharme-Barth (NOAA), Darcy Webber (Quantifish), Philipp Neubauer (Dragonfly)

  https://github.com/N-DucharmeBarth-NOAA/opal

SC22 project proposal

Moving from recommendations to software development

The plan is to deliver a WCPFC project proposal to SC22 in 2026, to follow up on recommendations from the scoping project

This will include a proposed budget, work plan, and timeline

Budget

From the scoping project report at SC21 in 2025

As an initial estimate, the development project could be launched as a three-year project from 2027 to 2030, with an annual budget of $200,000

This would allow:

$50,000 per year for workshops and smaller development subprojects, in the same manner as the current scoping project; and

$150,000 per year to divide between the development options that will be prioritized

Work plan

The scoping project is currently working with domain experts to draft a work plan for the project that will be proposed at SC22

One work plan for developing the software for tagging analysis and another work plan for developing the tuna assessment model software

Timeline

DTU Movement Model

Year	Activity	Amount required
2026	Demonstrate method at SC22 Contribute to SC22 Project Proposal	Small
2027	Enhance statistical method and software Deliver abundance indices for SKJ assmt	Medium
2028	Finalize SKJ abundance indicies Help SPC analyze YFT and BET indices	Medium
2029	Finalize YFT and BET indices	Small

Timeline

Tuna Assessment Model

Year	Activity	Amount required
2026	Develop prototype model components Demonstrate software at SC22 Contribute to SC22 Project Proposal	Medium
2027	Develop basic tuna assessment model Test model on single-region BET dataset	Large
2028	Develop prioritized model features Test model on SKJ dataset	Large
2029	Develop prioritized model features Test model on YFT and BET datasets	Large

Summary

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SC22 Project Proposal *budget, work plan, timeline*

These slides:

https://github.com/PacificCommunity/ofp-sam-yft-2026-assessment